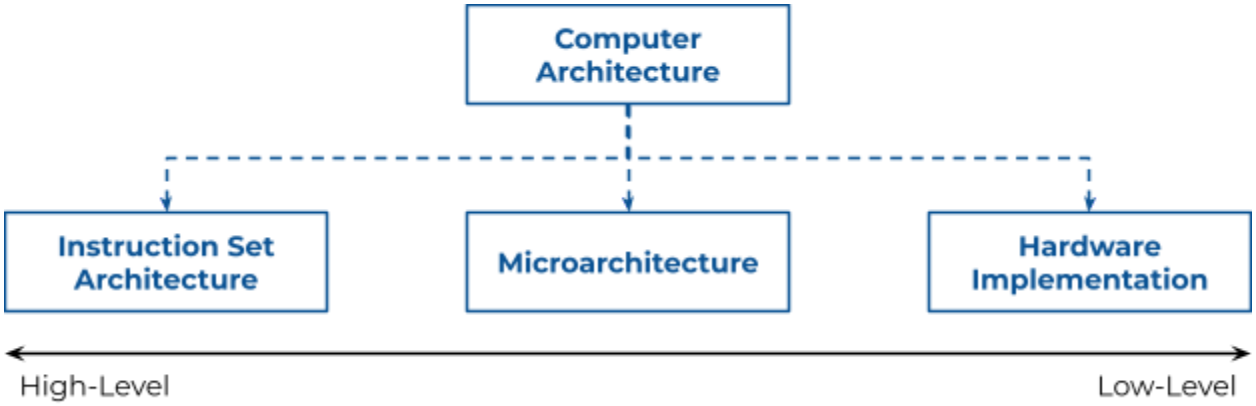
Introduction to Computer Organization



Computer Architecture

- Defines the conceptual structure and functional behavior of a computer system as seen by a programmer.
- Acts as the blueprint of a computing system, describing what the system should do rather than how it is implemented.

Instruction Set Architecture (ISA)

The part of computer architecture that is visible to the programmer. It **defines the instructions a processor can execute** and the way a programmer interacts with the system.

- The main components of the ISA include:
    - **Instruction Set**: The set of commands a processor understands, such as ADD, SUB, MUL, or LOAD.
        - **RISC (Reduced Instruction Set Computing)**: Simple and fixed-length instructions.
        - **CISC (Complex Instruction Set Computing)**: Complex and variable-length instructions.
    - **Addressing Modes**: How operands are accessed (immediate, direct, indirect, indexed).
    - **Registers**: Small, high-speed storage within CPU used for execution.
    - **Data Types**: Defines how numbers and characters are represented (integer, floating-point).

Computer Organization

Refers to the implementation details of the ISA. It determines **how the ISA is realized in hardware**, including the design of components like the processor, memory hierarchy, and control logic.

- The main components of computer organization include:
  - **Control Signals**: Timing and sequencing of operations.
  - **Datapath Design**: How data moves between ALU, registers, and memory.
  - **Memory Hierarchy**: Cache, main memory, and secondary storage design.
  - **I/O Organization**: Bus systems, device controllers, and interrupts.

Hardware Implementation

Refers to the **physical realization of the architecture**. It involves designing and fabricating the actual circuits, chips, and boards that make up the computer.

- The different levels of hardware implementation include:
  - **Digital Logic Level**: Logic gates (AND, OR, NOT) and flip-flops.
    - Example: Building a 1-bit full adder circuit.
  - **Microarchitecture Level**: Designing datapath, ALU, control unit.
    - Example: Implementing pipelining for faster instruction execution.
  - **System Level**: Integration of CPU, memory, and I/O subsystems on a motherboard.
    - Example: Multi-core processors with shared caches.